

Pierpaolo Vendittelli

Postdoctoral Researcher · AI for Digital Pathology

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EXPERIENCE

Postdoc Researcher

Prinses Maxima Centre for Pediatric Oncology

Dec. 2024 – Present

Utrecht, The Netherlands

- Developing deep-learning and MIL pipelines for WSI analysis of Wilms tumor, targeting anaplasia grading and clinical outcome prediction.
- Coordinating multi-center dataset integration across collaborating hospitals; enforcing reproducible, clinically validated computational workflows.
- Translating model outputs into interpretable biomarkers in close collaboration with pathologists and pediatric oncologists.
- Supervising students and contributing to peer-reviewed publications in computational pathology and pediatric oncology.

PhD Candidate – AI for Digital Pathology

Radboud University Medical Center

Mar. 2021 – Jun. 2026 (exp.)

Nijmegen, The Netherlands

- Developed deep-learning pipelines for pancreatic cancer WSI analysis within PANCAIM (EU Horizon 2020): multi-class U-Net for TSR quantification (368 patients / 4 datasets / 25 NL labs; Dice 0.75–0.86; TSR AUC 0.61 on TCGA-PAAD). Published in *PLOS ONE* (2024) and *SPIE MI* (2022).
- Extended work to multi-tissue segmentation for combining AI-based biomarkers (TSR, mitosis density, stromal cell density, TTR) in multivariate survival analysis across four independent PDAC cohorts. Published at *COMPAYL* (MICCAI 2025 workshop).
- Co-authored multi-center clinical studies: *European Radiology* (2025) on end-to-end prognostication; *The Lancet Oncology* (2025) PANORAMA study on AI-assisted pancreatic cancer detection; *Medical Image Analysis* (2025) scoping review on multimodal AI.
- Teaching Assistant for *Image Analysis* and *Medical Imaging Analysis* across 4 course editions at Radboud University; co-supervised Master's theses.

Machine Learning Engineer

Everis Italy (NTT DATA)

Oct. 2020 – Feb. 2021

Rome, Italy

- Built AI chatbots and conversational systems for clients in banking and automotive.
- Annotated industrial data for object recognition systems and developed live translation pipelines for video content.

Research Intern

ViCOROB – NeuroImage Computing Group, University of Girona

Feb. 2020 – Sep. 2020

Girona, Spain

- Benchmarked MRI synthesis architectures (ResU-Net, cGAN, CycleGAN) on WMH and BraTS 2018 datasets for T1↔FLAIR synthesis.
- Demonstrated ResU-Net achieves superior SSIM and lower MSE vs. GAN variants under data-scarce conditions; informed Master's thesis (grade 8.6/10).

PROJECTS & PUBLICATIONS

Social Agent

github.com/PierpaoloV/social-agent

Personal Project

2026 – Present

Human-in-the-loop social agent (Python, OpenAI API): captures private Telegram notes and GitHub milestones, ranks ideas, drafts post options, routes approvals via Telegram, and schedules publishing with a separate private state store and tests.

Weekly AI Paper-Triage Agent

github.com/PierpaoloV/weekly_ai_paper

Personal Project

2024 – Present

Multi-agent pipeline (Anthropic API, Python): weekly arXiv / PubMed scan → LLM summarisation → interest-based scoring (0–10) → automated email digest. In active personal use since 2024.

AI Job-Search Agent

github.com/PierpaoloV/job-agent

Personal Project

2025 – Present

End-to-end Python pipeline: Gmail API ingestion of job-alert emails → structured parsing (LinkedIn, Indeed, Glassdoor) → SQLite deduplication → LLM ranking with prompt caching (Claude Haiku) → Telegram digest. Runs 3×/day via GitHub Actions.

Automatic quantification of tumor-stroma ratio as a prognostic biomarker for pancreatic cancer

doi:10.1371/journal.pone.0301969

PLOS ONE

May 2024

Multi-class U-Net for TSR segmentation across 368 patients / 4 datasets (25 NL labs). Dice 0.75 (epithelium) and 0.86 (tumor bulk) on external TCGA-PAAD validation. TSR AUC 0.61 for 6-month survival prediction on independent cohort.

Automatic quantification of TSR as a prognostic marker for pancreatic cancer

openreview.net/pdf?id=Dtz_iaUpGc

MIDL 2023

Sep. 2023

Poster presentation: segmentation of tumor epithelium, healthy epithelium and TSR estimation from pancreatic WSIs.

Setting the research agenda for clinical AI in pancreatic adenocarcinoma imaging

doi:10.3390/cancers14143498

Cancers

Jul. 2022

Systematic review on the current role of AI for pancreatic cancer diagnosis and personalised treatment.

Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas

SPIE MI 2022

doi:10.1117/12.2611542

Apr. 2022

Multi-task U-Net with auxiliary classification head for coarse pancreatic tumour area segmentation in WSIs; classification head reduced false positives.

End-to-end prognostication in pancreatic cancer by multimodal deep learning: a retrospective, multicenter study

European Radiology

2025

Schuurmans et al. Multimodal deep learning combining CT imaging and clinical data for PDAC outcome prediction across multiple centers.

Artificial intelligence and radiologists in pancreatic cancer detection (PANORAMA): an international observational study

The Lancet Oncology

2025

Alves et al. International paired non-inferiority study comparing AI and radiologists for pancreatic cancer detection on standard-of-care CT.

Navigating the landscape of multimodal AI in medicine: a scoping review

Medical Image Analysis

2025

Schouten et al. Scoping review on technical challenges and clinical applications of multimodal AI in medicine.

Combining AI-based Biomarkers for PDAC Survival

COMPAYL (MICCAI 2025 Workshop)

Sep. 2025

Hensens, Sabroso-Lasa, Verbeke, Malats, Litjens, Vendittelli et al. Multi-tissue segmentation pipeline combining TSR, mitosis density, stromal cell density and TTR for multivariate PDAC survival prediction.

EDUCATION

Joint M.Sc. in Medical Imaging and Applications

Sep. 2018 - Sep. 2020

University of Burgundy, University of Cassino and Southern Lazio, University of Girona

France, Italy, Spain

- Final Grade: Bien (France), 102/110 (Italy), 8.6/10 (Spain)
- Scholarships: 2018-2020 Consortium Grant for the Erasmus Mundus Joint Master Degree.
- Relevant Courses: Computer-Aided Diagnosis, Medical Image Segmentation and Registration, Advanced Image Analysis.

B.Sc. in Informatics and Telecommunication Engineering

Sep. 2012 - Sep. 2017

University of Cassino and Southern Lazio

Cassino, Italy

- Final Grade: 94/110
- Scholarships: 2016 Erasmus+ Funded Exchange at Fontys University of Applied Sciences in Eindhoven, The Netherlands.
- Relevant Courses: Software Engineering, Databases, Mathematical Analysis, Object-Oriented Programming, Statistics.

Honors & Grants

2021 – **PhD Fellowship**, PANCAIM – Horizon 2020 EU-funded consortium for AI in pancreatic cancer

Nijmegen, The Netherlands

2018 – **Consortium Grant**, Erasmus Mundus Joint Master Degree in Medical Imaging and Applications (MAIA)

France, Italy, Spain

2016 **Erasmus+ Exchange Grant**, Funded exchange semester at Fontys University of Applied Sciences

Eindhoven, The Netherlands

Conference Presentations

Medical Imaging with Deep Learning (MIDL 2023)

Nashville, TN, USA

Poster Presentation

Sep. 2023

- “Automatic quantification of TSR as a prognostic marker for pancreatic cancer” – segmentation of tumor epithelium, healthy epithelium and tumor-stroma ratio from H&E WSIs.

SPIE Medical Imaging 2022

San Diego, CA, USA

Poster Presentation

Apr. 2022

- “Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas” – multi-task U-Net with classification head for pancreatic tumour area segmentation.

COMPAYL Workshop, MICCAI 2025

Daejeon, South Korea

Poster Presentation

Sep. 2025

- “Combining AI-based Biomarkers for PDAC Survival” – multi-tissue segmentation pipeline integrating TSR, mitosis density, stromal cell density and tumor-to-tissue ratio for multivariate survival prediction in pancreatic cancer.

- “Automatic classification of Anaplasia in Wilms Tumor” – deep learning pipeline for automated anaplasia grading in pediatric kidney tumor WSIs.

SKILLS

Languages Python, C++, C, Bash, SQL

ML / Deep Learning PyTorch, scikit-learn, HuggingFace, Transformers, MONAI, Keras

MLOps & Infrastructure Docker, Git, SLURM / GPU clusters, OpenSlide, W&B

LLM & Agents OpenAI API, Anthropic API, prompt engineering, multi-agent orchestration

CERTIFICATIONS

- **Machine Learning**, Stanford University
- **Convolutional Neural Networks**, deeplearning.ai
- **Generative Adversarial Networks (GAN) Specialization**, deeplearning.ai
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization & Optimization**, deeplearning.ai
- **Neural Networks and Deep Learning**, deeplearning.ai
- **AI for Medical Diagnosis**, deeplearning.ai
- **AI for Medical Prognosis**, deeplearning.ai
- **Applied Plotting, Charting & Data Representation in Python**, University of Michigan